Hence, the probability mass function of X is

$X = x$	0	1	2
$p(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Then, we can form a cumulative distribution function of X is

$X = x$	0	1	2
$p(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
$F_X(x) = P(X \le x)$	$\frac{1}{4}$	$\frac{1}{4}+\frac{1}{2}=\frac{3}{4}$	$\frac{3}{4}+\frac{1}{4}=1$

Example 6.6

A coin is tossed thrice. Let X be the number of observed heads. Find the cumulative distribution function of X.

Solution:

The sample space (S) ={ (HHH), (HHT), (HTH), (HTT),(THH),(THT),(TTH),(TTT)}

X takes the values: 3, 2, 2, 1, 2, 1, 1, and 0

Range of $X(R_x)$	0	1	2	3
$P_x(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$
$F_x(x)$	$\frac{1}{8}$	$\frac{4}{8}$	$\frac{7}{8}$	1

Thus, we have $$F_X(x)=\begin{cases}0, & for\ x<0\\ \frac{1}{8}, & for\ 0\le x<1\\ \frac{4}{8}, & for\ 1\le x<2\\ \frac{7}{8}, & for\ 2\le x<3\\ 1, & for\ x\le 3\end{cases}$$

Example 6.7

Construct the distribution function for the discrete random variable X whose probability distribution is given below. Also draw a graph of p(x) and F(x).

$X = x$	1	2	3	4	5	6	7
$P(x)$	0.10	0.12	0.20	0.30	0.15	0.08	0.05

Solution

From the values of $p(x)$ given in the probability distribution, we obtain

$$F(1) = P(x \le 1) = P(1) = 0.10$$

$$F(2) = P(x \le 2) = P(1) + P(2)$$
$$= 0.10 + 0.12 = 0.22$$

$$F(3) = P(x \le 3) = P(1) + P(2) + P(3)$$
$$= F(2) + P(3)$$
$$= 0.22 + 0.20$$
$$= 0.42$$

$$F(4) = F(3) + P(4)$$
$$= 0.42 + 0.30$$
$$= 0.72$$

$$F(5) = F(4) + P(5)$$
$$= 0.72 + 0.15$$
$$= 0.87$$

$$F(6) = F(5) + P(6)$$
$$= 0.87 + 0.08$$
$$= 0.95$$

$$F(7) = F(6) + P(7)$$
$$= 0.95 + 0.05$$
$$= 1.00$$

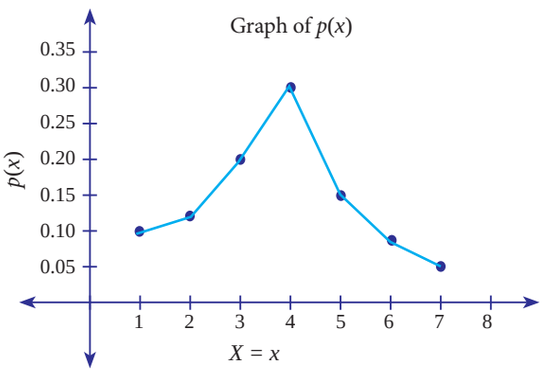


Fig.6.3

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